



Jixels ERP System Overview User Manual

A practical guide for Director, Admin, Branch, Agent, HR, Finance, Operations, and Sales users.

Version

Latest build,
May 2026

Deployment

GitHub Pages /
Vercel ready

Storage now

IndexedDB
shared browser
store

Database later

API/KV layer
ready for server
database





System Overview

How the ERP works as one unit

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The system is a portal hub for phone sales, branch inventory, employee payroll, finance reporting, approvals, and management oversight.

Director

Reads company-wide data, branch performance, reports, stock, sales, and department summaries.

Admin

Approves accounts, approves branches/agents, exports data, deletes data, and generates reports.

Branch

Adds stock, records sales, records credit sales, tracks damage/loss, and exports branch reports.

Agent

Sells phones assigned to the branch, records full payment or credit sale, and triggers payment flow.

Departments

HR, Finance, Operations, and Sales read the same ERP data for payroll, money, stock, and sales reports.

PWA

The site can be installed as an app and includes a manifest, service worker, icon, logo, and favicon.

Data Flow

- 1 Admin or Branch creates branch/agent/department accounts and approvals.
- 2 Branch adds phone inventory, including serial, model, color, storage, and price.
- 3 Agent or Branch completes sale as full payment or sell on credit.
- 4 ERP updates stock, sold phones, finance totals, transaction history, SMS queue, and reports.
- 5 Director and departments view monthly/yearly reports from the same shared data.



Accounts and Approvals

Registration and access control

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Department Accounts

Users register as HR, Finance, Operations, Sales, or Admin. Login uses the same registered credentials after approval rules are satisfied.

Branch Accounts

Each branch can register once. Admin approves new branches before normal use.

Agent Accounts

Each agent registers once and is restricted from registering in other portals once already registered.

Director Account

Director login is kept separate and can view department portals in read-only mode.

Approval Rules

Account	Approval	Result After Approval
Branch	Admin approves branch registration.	Branch login opens the Branch Dashboard.
Agent	Admin approves agent registration and branch assignment.	Agent login opens the Agent Dashboard.
Departments	Admin controls department access and records.	Departments login opens the correct role dashboard.

Good Login Practice

- Use the same username and password created during registration.
- Check approval status if a new branch or agent cannot open.
- Use Admin to approve pending accounts and confirm approval success messages.
- Director can view data but should not change department records from read-only access.



Director and Admin

Control, oversight, and reporting

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Director Dashboard

- Realtime overview of branches, stock, sold phones, and employees.
- Inventory by branch with search.
- Company weekly, monthly, yearly, and general reports.
- Reads department portals without changing department data.

Admin Dashboard

- Approves branches, agents, and department accounts.
- Views every account and approval status.
- Exports all ERP/HR/account/audit/SMS data as JSON.
- Can delete all data after confirmation.

Admin Data Controls

Feature	Purpose	Safety Note
Export JSON	Downloads all current ERP data for backup or migration.	Keep exports private because they contain system records.
Delete all data	Clears ERP, HR, accounts, audit, notification, and SMS queues.	Requires the confirmation phrase before running.
Approval panel	Approves pending branch and agent accounts.	Approved account shows success status before login.
Reports	Admin can generate monthly and yearly reports.	Reports include paid sales, credit phones, and credit balance.



Branch Portal

Inventory, sales, credit, and reports

Inventory Entry

Branch adds phones by serial number, model, color, storage, and price. Duplicate branch registration is restricted.

Sales Completed

Branch can complete a sale directly or view sales completed by agents from the same branch.

Damage and Loss

Branch records damaged and lost stock for operations and Director reporting.

Branch Reports

Branch generates reports, CSV, Word, and print/PDF output with monthly/yearly period support.

Sell on Credit in Branch

1. Select the phone serial number.
2. Enter customer phone number.
3. Select channel: M-Pesa or Bank.
4. Select sale type: Full payment or Sell on credit.
5. For credit, enter amount paid and credit due date.
6. Complete sale. The system saves amount, paid, balance, due date, and status.

Tracked Field	Meaning
Amount	Full phone selling price.
Paid	Cash collected now; this is what finance totals count as received money.
Balance	Outstanding credit amount.
Due date	Date customer should clear the credit balance.
Status	Completed, Credit due, or Credit cleared.



Agent Portal

Selling phones and credit tracking

Agents use approved accounts and sell phones for their assigned branch. Agent sales immediately update branch inventory, sold phones, finance totals, SMS queue, and reports.

Full Payment Sale

- Choose serial and customer phone.
- Choose M-Pesa or Bank.
- Amount comes from phone price or can be entered.
- System records completed sale.

Sell on Credit

- Select Sell on credit.
- Enter amount paid now.
- Enter due date.
- System tracks paid amount and remaining balance.

Agent Sales History

Column	Use
Date	When sale was recorded.
Model / Serial	Phone sold and unique serial identifier.
Customer	Customer phone number for payment/SMS tracking.
Channel / Reference	M-Pesa or Bank and transaction reference.
Amount / Paid / Balance	Full price, money received now, and credit still outstanding.
Status	Completed, Credit due with due date, or Credit cleared.

Important Rule

After a phone is sold, including credit sale, it is removed from available inventory and appears in sold phones so the same serial cannot be sold again.



Departments

HR, Finance, Operations, and Sales

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HR

Adds employees, assigns branches, manages salary, debt type, debt amount, and forwards payroll to Finance.

Finance

Reviews payroll, accepts/rejects payroll items, records paid payroll, and reads received money totals.

Operations

Views stock, sold phones, damaged/lost records, and operational reports by branch.

Sales

Views sales activity, period sold, paid revenue, credit phones, and credit balance across branches.

HR Payroll by Branch

- Use View branch with entries like 1, 01, b01, or Branch 01.
- If no branches exist, the page shows: There is no branch registered yet.
- If branch exists but has no employees, it shows no active employees found.
- Forward branch sends that branch payroll queue to Finance.
- Forward all sends all active employees to Finance.

Finance Review

Action	Result
Accept payroll item	Marks Finance review as accepted and allows payment.
Reject payroll item	Blocks payment until accepted later.
Pay item or Pay all	Marks payroll as paid and sends notifications to Finance/HR.



Reports and Exports

Monthly, yearly, Word, PDF, CSV, and JSON

Reports are available in Director, Admin, Branch, HR, Finance, Operations, and Sales areas. They support monthly and yearly generation where applicable.

Portal	Report Output	Includes Credit Sales
Director	Company general, weekly, monthly, yearly, branch reports, CSV, Word, PDF.	Yes: credit phones and credit balance.
Admin	Inventory report, monthly/yearly summaries, JSON export, PDF/Word.	Yes: credit sales in period.
Branch	Branch operations report, transactions CSV, Word, PDF.	Yes: amount, paid, balance, sale type, due date, status.
Agent	Sales history CSV.	Yes: amount, paid, balance, credit due date, status.
Finance	Branch finance report and payroll queue report.	Yes: credit phones and outstanding balance.
Sales / Operations	Inventory, sales, stock, damaged/lost, monthly/yearly reports.	Yes: period credit phones and credit balance.

Export Rules

- CSV exports are used for spreadsheet review.
- Word downloads are used for editable office documents.
- Print/PDF creates printable reports from the browser.
- Admin JSON export is the full backup/migration file.
- Reports use paid amount as received revenue and keep credit balance separate.



Integrations, PWA, and Data

Ready for Vercel, Daraja, and OneSignal

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IndexedDB Now

The browser uses IndexedDB/KV client so pages read the same local data instead of separated localStorage-only records.

Database Later

The KV/API layer is ready to connect to a real database later so the same ERP structure can sync online.

M-Pesa Daraja

STK Push endpoint is ready for Vercel. Add Safaricom Daraja keys in Vercel environment variables.

OneSignal

Notification and SMS endpoints are prepared for OneSignal keys in Vercel environment variables.

PWA

The app has manifest, service worker, install button, logo, favicon, and icon support for installable app behavior.

GitHub Pages

The frontend is deployed from the main branch. Vercel can host APIs for server integrations.

Transaction Notifications

- When transaction occurs, Finance, Sales, Operations, Branch/Admin, and Director notification audiences can be targeted.
- Customer SMS is queued after sale or credit sale.
- For M-Pesa credit sale, STK push uses only the amount paid now.
- Full sale value and unpaid balance stay in transaction history for tracking.



Security Overview

CIA, ACID, audit, and operational safety

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Confidentiality

Role-based access keeps users in their correct portal. Director read-only mode limits department changes. Exports should be shared only with authorized people.

Integrity

Serial numbers prevent duplicate phone sales. Credit sales store full amount, paid amount, balance, due date, and status for accurate records.

Availability

PWA and IndexedDB allow local use. Vercel/API/database connection can improve multi-device availability later.

Audit

Important actions such as login, approvals, reports, payroll actions, SMS queue, and errors are audit-ready.

Transaction Safety and ACID Direction

Property	How the ERP Supports It
Atomicity	Sale actions update transaction log, sold phone record, inventory, finance summary, and SMS queue together in one flow.
Consistency	Reports use the same transaction fields across Agent, Branch, Director, Admin, Finance, Sales, and Operations.
Isolation	Role pages separate user actions. Future database should enforce server-side locks/transactions for multi-user writes.
Durability	IndexedDB keeps local records. Admin JSON export and future database sync improve long-term durability.

Recommended Security Practice

- Use strong passwords for Director and Admin accounts.
- Approve only known branches and agents.
- Keep Daraja and OneSignal keys only in Vercel environment variables, never inside public frontend files.
- Back up Admin JSON exports before using Delete all data.
- When connecting a real database, enforce server-side authentication, authorization, and transaction validation.